Shaft end and sleeve's axial thrust

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1. Purpose & Scope

This procedure describes how to calculate the axial thrust contribution of sleeves and shaft ends.

These contributions shall be considered during the calculation of the total axial thrust acting on the pump's rotor.

This procedure is applicable to all the centrifugal pumps, when it is required to calculate the acting axial forces of shaft sleeves and shaft ends

This procedure is applicable to Nova Milanese and La Spezia plants.

2. Abbreviations

DE Drive End side

NDE Non-Drive End side

PLB Pumped Liquid Bearing

3. General procedure: description

Shaft ends and rotor sleeves may be exposed to high pressures or forces acting on their surfaces. Hence, these contributions may generate a significant resultant axial thrust.

4. Calculation method

The calculation of the resultant axial thrust acting on shaft ends or sleeves is calculated based on the following formula:

$$F = \frac{\pi}{4} \cdot \left[\left(D_{2e}^2 - D_{2i}^2 \right) \cdot p_2 - \left(D_{1e}^2 - D_{1i}^2 \right) \cdot p_1 \right]$$

Where:

- D_{1i} [m] and D_{1e} [m] are respectively the internal and the external diameters of the sleeve on side 1.
- D_{2i} [m] and D_{2e} [m] are respectively the internal and the external diameters of the sleeve on side 2.
- F [N] is the resultant force acting on the sleeve or bush in subject (> 0, if directed towards the pump suction).
- p_1 [Pa] and p_2 [Pa] are respectively the fluid static pressures on sides 1 and 2 acting on the sleeve's surfaces. These pressures shall be duly calculated based on the actual sleeve type and operating conditions.

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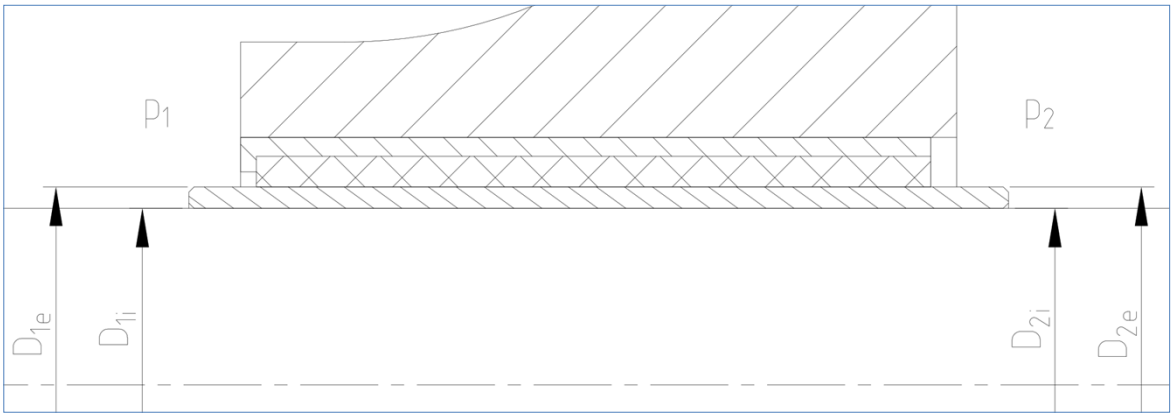

Figure 4.1: Reference drawing of a typical shaft sleeve.

4.1. Interstage sleeve

The interstage sleeve is installed between consecutive impellers. The axial thrust is calculated considering the formula from Section 4. Different conditions occur based on the actual installation:

1) If the interstage sleeve is installed between consecutive single suction impellers (refer to *Figure 4.2*), p_1 is the previous stage impeller's differential pressure reduced by the centrifugal effect on the rear shroud (refer to **SOP-435** and **Mod.530** for further details) and p_2 is the suction pressure of the next stage impeller (equal to the total differential pressure of the previous stage, downstream the volute or the diffuser).

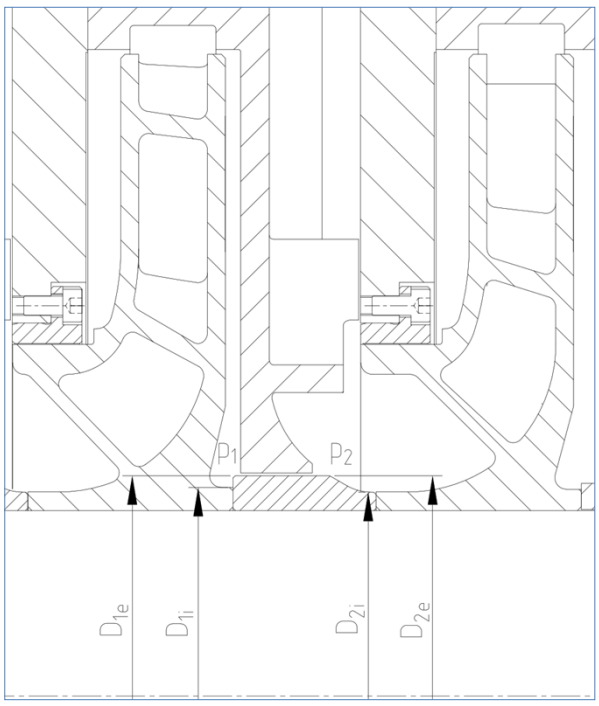

Figure 4.2: Interstage sleeve between consecutive impellers in horizontal pumps.

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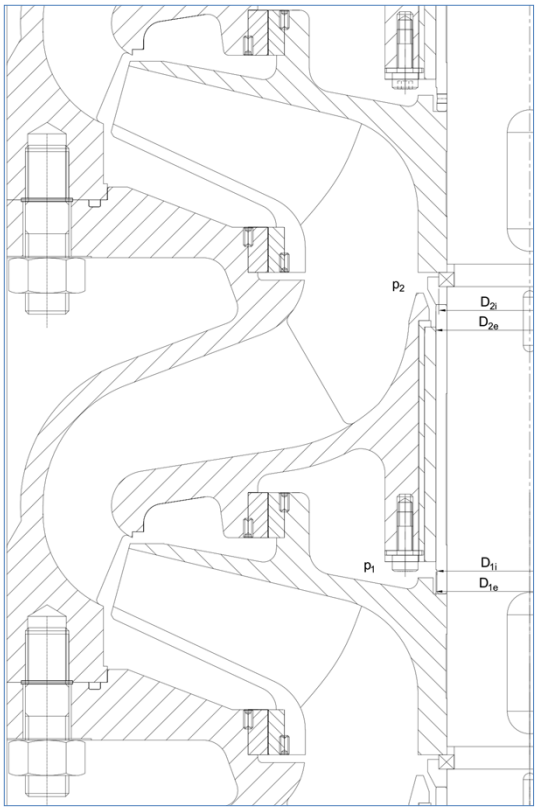

Figure 4.3: Interstage sleeve between consecutive impellers in vertical pumps.

2) If the interstage sleeve is installed downstream a double suction impeller (refer to *Figure 4.4*), p_1 is the pump's suction pressure and p_2 is the suction pressure of the next stage impeller (equal to the total differential pressure of the previous stage, downstream the volute or the diffuser).

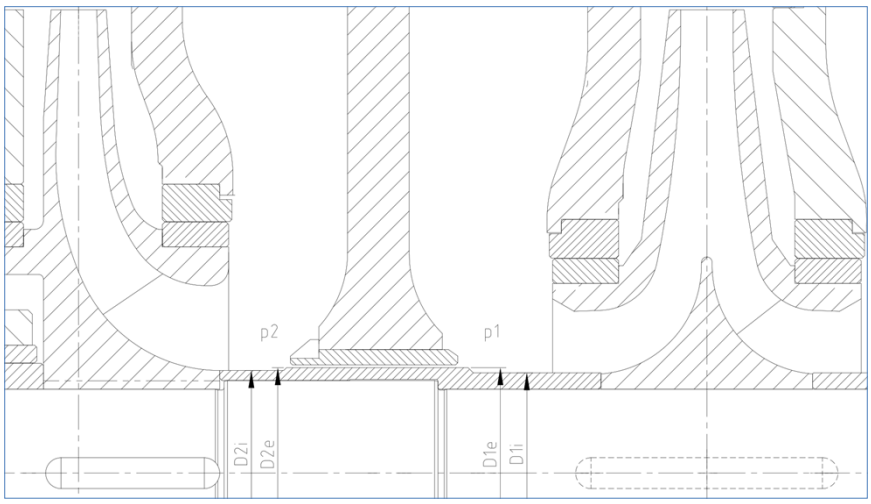

Figure 4.4: Interstage sleeve between 1st stage (double suction) and 2nd stage (single suction) impellers.

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4.2. Central bush

The central bush is installed in case of multi-stage back-to-back pumps. On this component, differential pressure acts and generates an axial thrust. The pressures acting on the sleeve's sides are calculated as the delivery pressures of the last in-line stage's and the last overall stage's impellers, reduced based on the pressure loss along the rear shrouds. The pressure drops along the impellers' shrouds are calculated according to J. F. Gulich's Centrifugal Pumps. There are two different calculation approaches, a simplified one and a complete one (refer to **SOP-435** for further details).

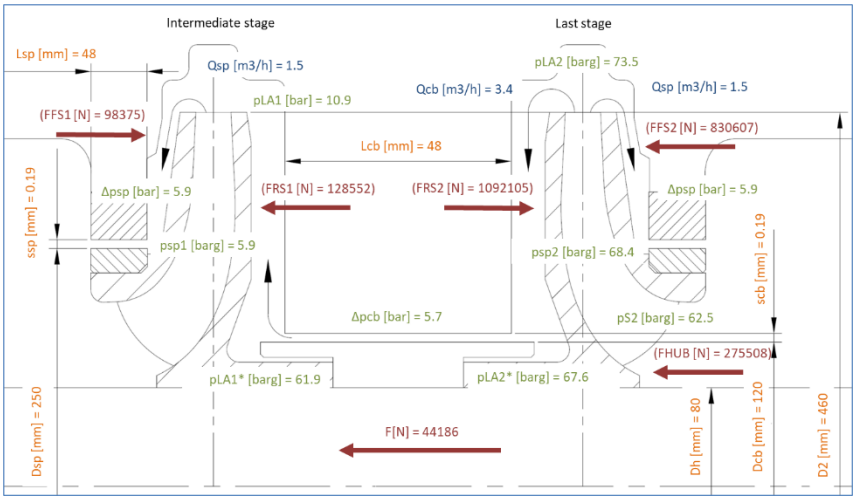

*Figure 4.5: Typical central bush arrangement (extracted from **Mod. 530**).*

In this case, the equation at section 4 is applied considering $p_2 = p_{LA2}^*$ and $p_1 = p_{LA1}^*$, which shall be calculated as per **Mod.530**.

4.3. End bush

The end bush sleeve is installed to isolate the stuffing box from the impellers' side. The axial thrust is calculated by means of the formula from Section 4, considering p_1 equal to the stuffing box pressure depending on the seal flushing type and p_2 equal to the internal pressure. The latter shall be defined based on the pump arrangement:

1) In case of end bush in the pump's suction side, it is equal to the pump's suction pressure (refer to *Figure 4.6)*.

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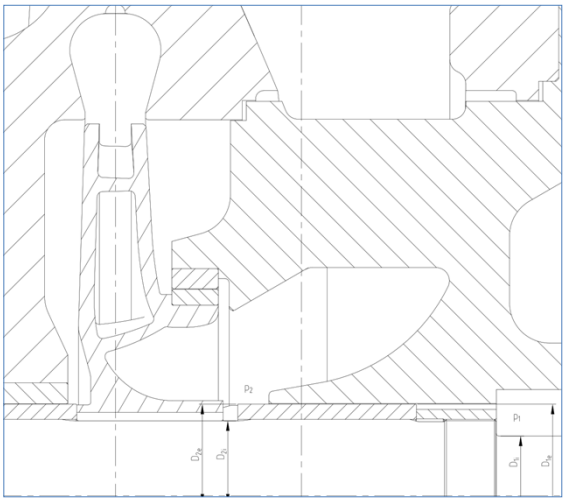

Figure 4.6: End bush in pump suction side.

2) In case of end bush installed in the pump's discharge side, the following cases can be identified:
   a. In case of in-line pumps, it is equal to the last stage impeller's discharge static pressure reduced by the loss along the rear shroud or to the overpressure in the balancing holes' chamber (if any). Refer to **SOP-435** and **Mod.530** for further details.
   b. In case of back-to-back pumps, it is equal to the suction pressure of the first back-to-back stage impeller (refer to *Figure 4.7).*

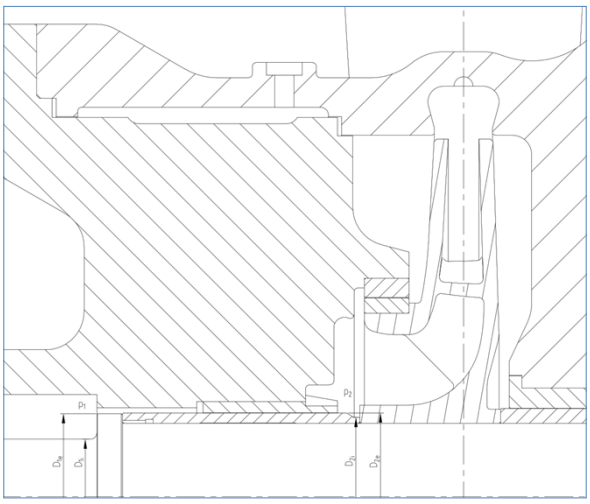

Figure 4.7: End bush in pump discharge side.

4.4. Mechanical seal sleeve

The same considerations as per previous sections are applicable for the mechanical seal's sleeve as well. In this case, the following main forces acting on the sleeve are identified:

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1) Internal pressure load: The pressure of the stuffing box acts on the exposed surface of the mechanical seal. This generates an axial thrust acting on the shaft sleeve, considering that its external surface is exposed to the atmospheric pressure.
2) Spring load: The preload and the operating load generated by the spring pushing together the sealing faces.
3) Buffer or barrier fluid pressure load: In case of double mechanical seal, the chamber could be pressurized by the buffer or barrier fluid.

In general, this axial thrust is not significant and not considered, since the surfaces exposed to the pressures are not wide.

In addition, to evaluate the actual axial thrust, all the data relevant to the mechanical seal's geometry shall be known. Unfortunately, this information is not commonly shared by the seal manufacturers. For this reason, it is recommended to ask them for the actual resultant axial force acting on the shaft in the operating conditions.

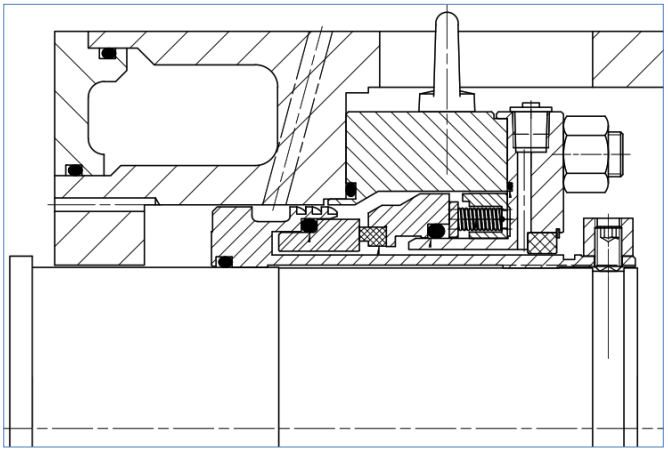

Figure 4.8: Reference drawing of single mechanical seal.

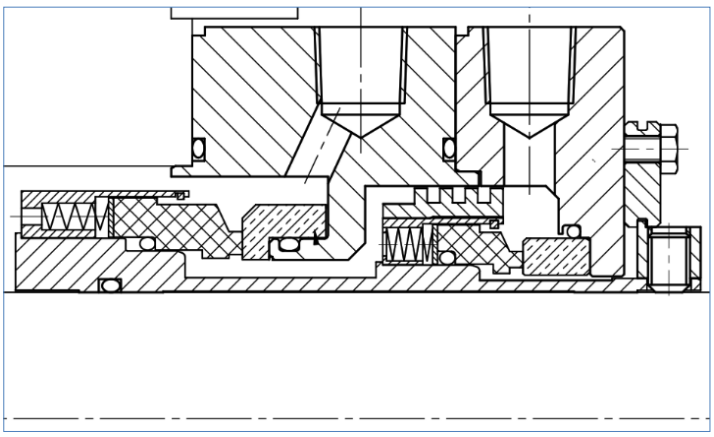

Figure 4.9: Reference drawing of double mechanical seal.

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4.5. Shaft end

In case that the shaft end is exposed to the pumped fluid, the fluid pressure generates an axial thrust on the pump's rotor. In this case, the formula from Section 4 is modified as indicated below:

$$F = \frac{\pi}{4} \cdot D^2 \cdot p$$

Where:

- D [m] is the average diameter of the mechanical seal faces.
- p [Pa] is the fluid static pressure acting on the shaft end.

This case is typical for the following designs:

1) In case of overhung impellers (such as in OH and VS type pumps as per API 610 standard), the pressure acts in the impeller's eye, generating an axial thrust (refer to Figure 4.10).

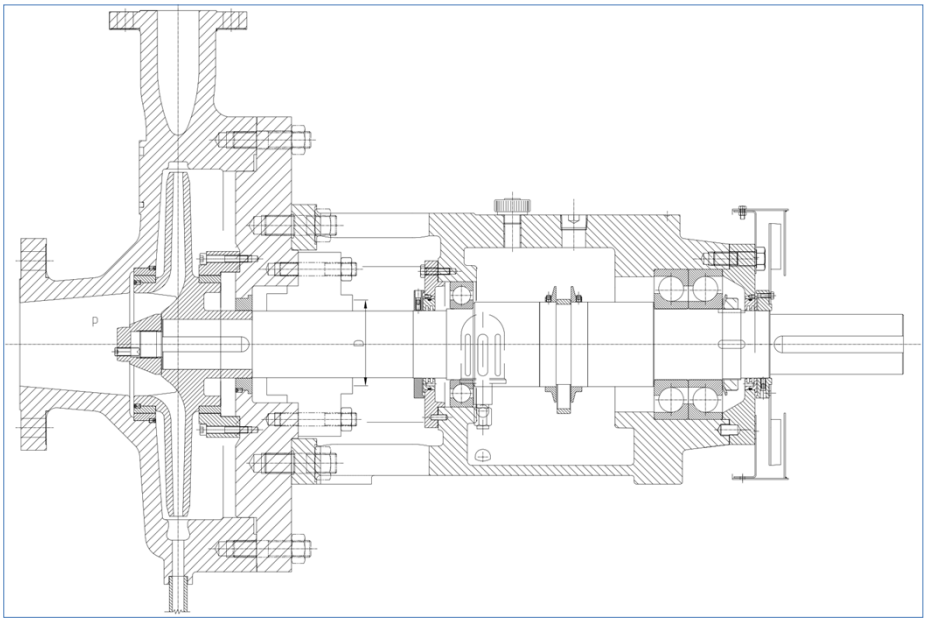

Figure 4.10: OH2 pump's sectional drawing.

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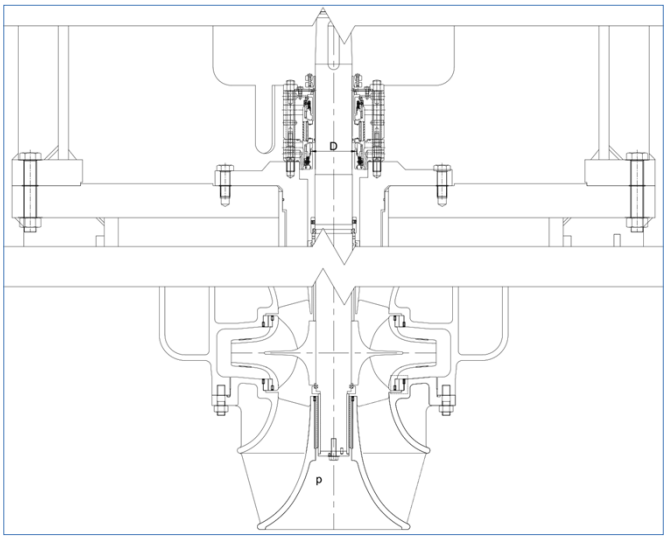

Figure 4.11: VS pump's sectional drawing.

2) The horizontal pumps equipped with PLBs (refer to *Figure 4.12*) are normally designed to avoid the installation of the mechanical seal in the NDE side as opposed to typical oil-lubricated bearings. In this case, the NDE shaft end is confined inside the pump's casing and it is in direct contact with the pumped fluid. For this reason, the fluid pressure generates an axial thrust acting on the surface of the shaft end.

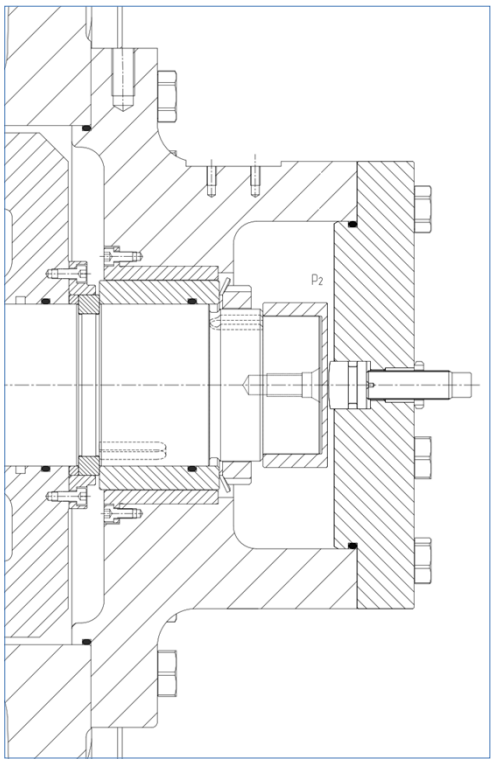

Figure 4.12 Pump equipped with PLB.

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5. References

Centrifugal Pumps, J. F. Gulich, 4th Edition, 2020

Mod.530 - *Impeller's hydraulic axial thrust*

SOP-435 - *Impeller's hydraulic axial thrust*

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